

Cybersecurity Report: Google Fitbit Air

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Figure 1: Google Fitbit Air

1 Physical Aspects

The Google Fitbit Air is designed as a minimalist health tracker. Below is a detailed breakdown of its physical specifications:

- **Form Factor:** A small, screenless "pebble" that houses the sensors and battery. It is designed to be completely unobtrusive.
- **Weight:** The sensor module weighs 5.2 grams. With a standard band attached, the total weight is approximately 12 grams.
- **Dimensions:** The sensor module measures 34.9mm in height, 17mm in width, and 8.3mm in thickness.

- **Physical Interface:** The device is entirely screenless. It features a single LED status light. Interaction is handled via physical gestures; for example, a firm double-tap wakes the LED to check the battery status.
- **Battery & Charging:** It has a 34.5 mAh rechargeable lithium-polymer battery and provides up to eight days of battery life.
- **Ports & Modularity:** There are no exposed USB ports. The pebble pops directly in and out of the wristbands and charges via a dedicated base.

2 IoT Architecture

The Fitbit Air functions transforms raw physiological data into actionable health insights. The architecture is broken down into the following six pillars:

2.1 Identification

- **Device:** The tracker uses a unique hardware identifier (MAC address) for secure pairing with a single smartphone, preventing interception of biometric data.
- **User/Service:** Identity is managed via the user's Google Account, ensuring the physical tracker is securely linked to a personal cloud database through the Google Health app.

2.2 Sensing

- **Sensors:** Optical sensors capture physiological metrics like heart rate and SpO2. Motion sensors track physical movement, orientation, steps, and sleep patterns.
- **Actuators:** A haptic vibration motor is used for silent alarms, alongside the LED for visual hardware status.

2.3 Networking

Communication relies on Bluetooth Low Energy to continuously pair and sync data with the user's smartphone. The smartphone functions as the local network gateway, leveraging standard internet protocols (Wi-Fi or Cellular) to transmit the localized data securely to Google's cloud infrastructure.

2.4 Computation

- **Resource-Constrained Node (The Tracker):** Handles edge-level processing with ultra-low power consumption. It buffers raw sensor data, recognizes physical tap gestures, and runs baseline automatic activity detection.
- **Gateway (Smartphone):** The Google Health app processes incoming BLE packets, formats the data for the local user interface, and routes it to the cloud.
- **Cloud (Google Servers):** Manages heavy computational workloads, stores long-term historical data, and executes resource-intensive AI models.

2.5 Services

The system exposes few functionalities to the user, including setting recurring silent alarms and requesting on-demand battery status via LED cues.

2.6 Analytics

Raw sensor data is aggregated and transformed into actionable intelligence on the back-end. Statistical models generate sleep stage scores. Additionally, generative AI algorithms (via the Google Health Coach) synthesize these biometrics to deliver customized weekly workout plans and recovery recommendations.